

■ 3.4.4 Getting Full Solutions

If you have an equation like $2x == 0$, it is perfectly clear that the only possible solution is $x \rightarrow 0$. However, if you have an equation like $ax == 0$, things are not so clear. If a is not equal to zero, then $x \rightarrow 0$ is again the only solution. However, if a is in fact equal to zero, then any value of x is a solution.

`Solve` implicitly assumes that the parameter a does not have the special value 0.

```
In[1]:= Solve[ a x == 0 , x ]
Out[1]= {{x -> 0}}
```

`Roots` makes the same assumption.

```
In[2]:= Roots[ a x == 0 , x ]
Out[2]= x == 0
```

`Reduce`, on the other hand, gives you all the possibilities, without assuming anything about the value of a .

```
In[3]:= Reduce[ a x == 0 , x ]
Out[3]= a != 0 && x == 0 || a == 0
```

The results that `Reduce` gives are logical statements representing all possible solutions to an equation, allowing for special values of parameters.

A result like $a \neq 0 \ \&\& \ x == 0 \ || \ a == 0$ is interpreted as follows. The `&&` stands for AND; `||` stands for OR. Since in *Mathematica* the `||` operator has lower precedence than `&&`, the expression is equivalent to $(a \neq 0 \ \&\& \ x == 0) \ || \ (a == 0)$. The two pieces then represent alternative solutions. The first alternative is that a is not equal to 0, and x equals 0. This is the solution that `Solve` finds. The second alternative is that a is equal to 0, but there is no restriction on x .

$lhs == rhs$	an equation asserting that two quantities are equal
$lhs != rhs$	an inequation asserting that two quantities are unequal
$eqn_1 \ \&\& \ eqn_2$	equations that must simultaneously be satisfied
$eqn_1 \ \ eqn_2$	alternative equations, only one of which need be satisfied

Logical forms associated with equations.

This is the solution to an arbitrary linear equation given by `Roots` and `Solve`.

```
In[4]:= Roots[a x + b == 0 , x]
Out[4]= x == - $\frac{b}{a}$ 
```

`Reduce` gives the full version, which includes the possibility $a=b=0$.

```
In[5]:= Reduce[a x + b == 0, x]
```

```
Out[5]= a != 0 && x == -( $\frac{b}{a}$ ) || a == 0 && b == 0
```

Here is the full solution to a general quadratic equation. There are three alternatives. If a is nonzero, then there are two solutions for x , given by the standard quadratic formula. If a is zero, however, the equation reduces to a linear one. Finally, if a , b and c are all zero, there is no restriction on x .

```
In[6]:= Reduce[a x^2 + b x + c == 0, x]
```

```
Out[6]= a != 0 && (x ==  $\frac{-\left(\frac{b}{a}\right) + \text{Sqrt}\left[\frac{b^2}{a^2} - \frac{4c}{a}\right]}{2}$  ||  

 $\frac{-\left(\frac{b}{a}\right) - \text{Sqrt}\left[\frac{b^2}{a^2} - \frac{4c}{a}\right]}{2}$ ) ||  

b != 0 && a == 0 && x == -( $\frac{c}{b}$ ) ||  

a == 0 && b == 0 && c == 0
```

The most important difference between `Reduce` and `Solve` is that `Reduce` gives *all* the possible solutions to a set of equations, while `Solve` gives only the *generic* ones. Solutions are considered “generic” if they involve conditions only on the variables that you explicitly solve for, and not on other parameters in the equations. `Reduce` and `Solve` also differ in that `Reduce` always returns combinations of equations, while `Solve` gives results in the form of transformation rules.

<code>Solve[eqns, vars]</code>	find generic solutions to equations
<code>Reduce[eqns, vars]</code>	reduce equations, maintaining all solutions

Solving equations.